

b Origins of the MSE

Training-shot
effects



Training bias

Training shot noise

$$N^{-2}$$



Training fluctuations

Training shot noise

$$1 / (N n_{\text{tr}})$$

Target noise
Classical



Training target noise

Noisy training labels

$$\gamma^2 / n_{\text{tr}}$$



Test target noise

Noisy test labels

$$\gamma^2$$

Test-shot
effect



Test sampling

Finite test shots

$$1 / M$$

Quantum
overfitting



Quantum overfitting

Training-shot noise

× target noise × test-shot noise

$$\gamma^2 N / M$$